



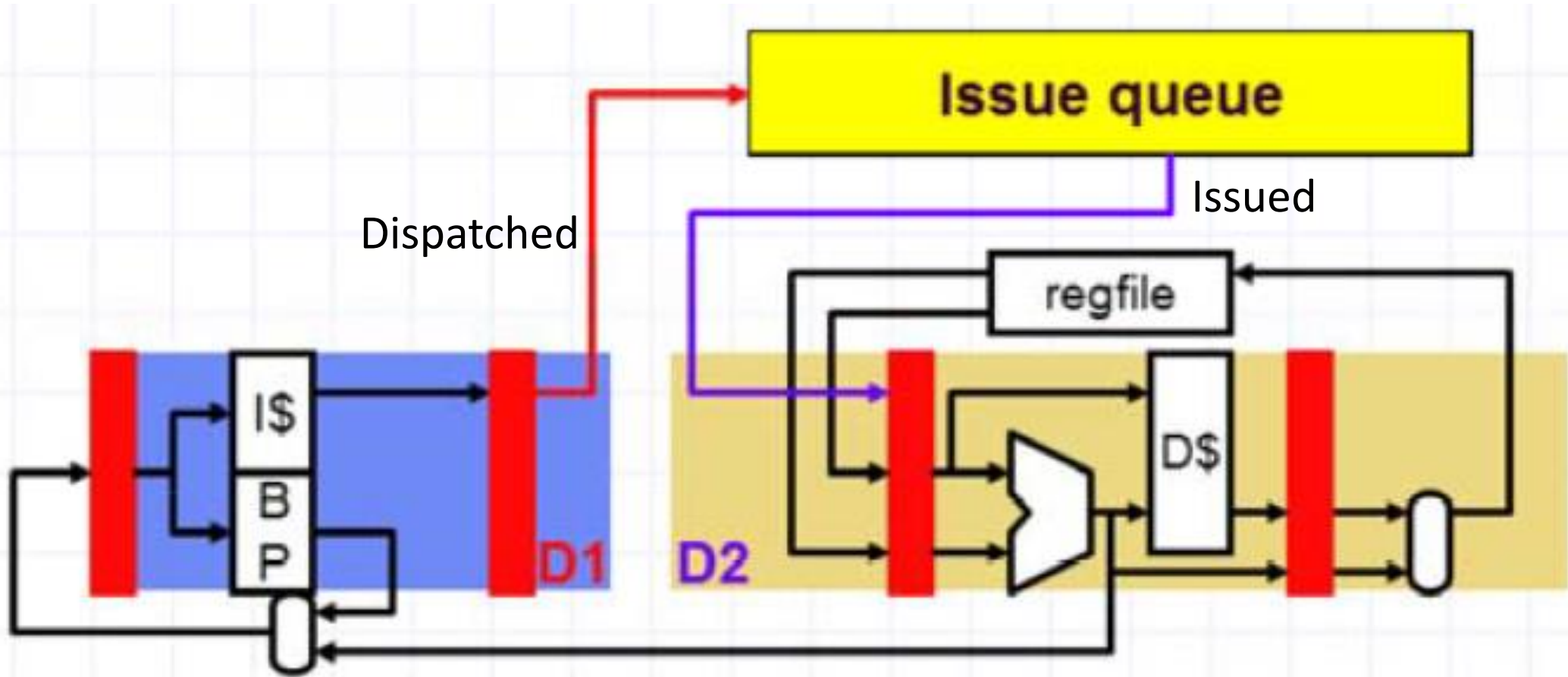
CS683: Advanced Computer Architecture

Lecture 19: Everything is out-of-order, except

<https://www.cse.iitb.ac.in/~biswa/courses/>

<https://www.cse.iitb.ac.in/~biswa/>

Issue Q (instructions enter in-order, leave out-of-order)



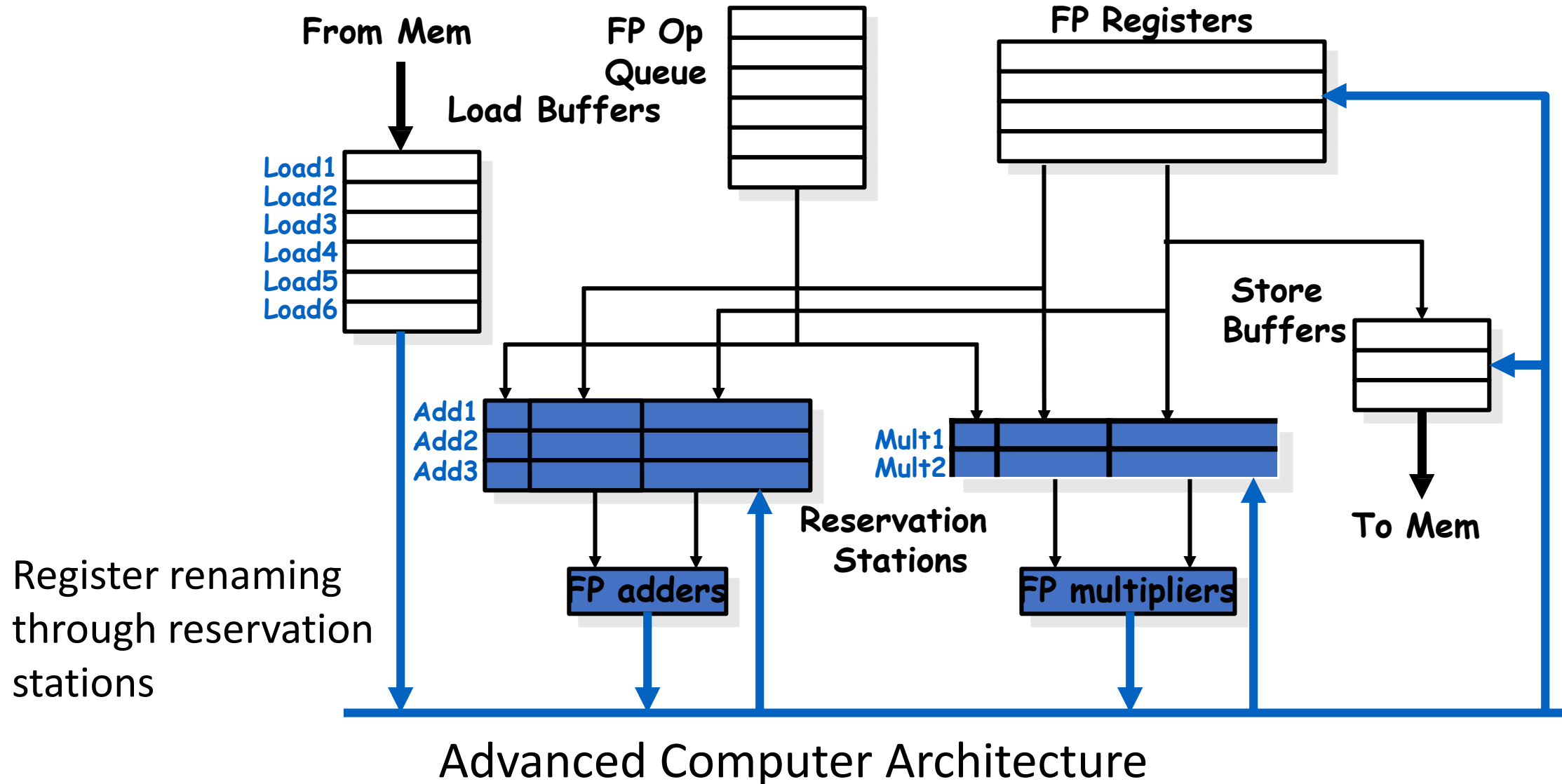
Why Issue Q or Operation Q?

In-order pipeline: younger ones cannot pass/cross the older instructions

Issue Q: Structure that can hold multiple instructions in-flight, actually not a Q

Register renaming also happens at this stage. More later 😊

The Complete Picture First: IBM 360



What is the deal?

- Control & buffers [distributed](#) with Function Units (FU);
 - FU buffers called “[reservation stations](#)”; have pending operands
- Registers in instructions replaced by values or pointers to reservation stations(RS); called [register renaming](#) ;
 - avoids WAR, WAW hazards
 - More reservation stations than registers, so can do optimizations compilers can't
- Results to FU from RS, [not through registers](#), over [Common Data Bus](#) that broadcasts results to all FUs
- Load and Stores treated as FUs with RSs as wells

Reservation Station

Op: Operation to perform in the unit (e.g., + or −)

V_j, V_k: Value of Source operands

- Store buffers has V field, result to be stored

Q_j, Q_k: Reservation stations producing source registers (value to be written)

- Q_j, Q_k=0 => ready
- Store buffers only have Q_i for RS producing result

Busy: Indicates reservation station or FU is busy

Register result status—Indicates which functional unit will write each register, if one exists. Blank when no pending instructions that will write that register.

Pipeline Stages++

1. Dispatch and Issue—get instruction from Op Queue

If reservation station free (no structural hazard),
control issues instr & sends operands (renames registers).

To make it simple for the rest of the lecture, let's assume only issue

2. Execution—operate on operands (EX) When both operands ready then execute; if not ready, watch Common Data Bus for result

3. Write result—finish execution (WB)

Write on Common Data Bus to all awaiting units; mark reservation station available

- Normal data bus: data + destination (“go to” bus)
- Common data bus: data + source (“come from” bus)
 - 64 bits of data + 4 bits of Functional Unit source address
 - Write if matches expected Functional Unit (produces result)
 - Does the broadcast

Example: Ready Steady Go!

Instruction status:

				Exec	Write
Instruction		<i>j</i>	<i>k</i>	Issue	Comp Result
LD	F6	34+	R2		
LD	F2	45+	R3		
MULTD	F0	F2	F4		
SUBD	F8	F6	F2		
DIVD	F10	F0	F6		
ADDD	F6	F8	F2		

	Busy	Address
Load1	No	
Load2	No	
Load3	No	

Load: 2 cycle
FP add: 2 cycles
FP multiply: 10 cycles
FP divide: 40 cycles

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
Time	Name	Busy	Op	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	No					
	Mult2	No					

Register result status:

Clock	<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
0	FU								

Cycle-1

Instruction status:

				Issue	Exec	Write
Instruction		<i>j</i>	<i>k</i>	Comp	Result	
LD	F6	34+	R2	1		
LD	F2	45+	R3			
MULTD	F0	F2	F4			
SUBD	F8	F6	F2			
DIVD	F10	F0	F6			
ADDD	F6	F8	F2			

	Busy	Address
Load1	Yes	34+R2
Load2	No	
Load3	No	

Load: 2 cycle
FP add: 2 cycles
FP multiply: 10 cycles
FP divide: 40 cycles

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
Time	Name	Busy	Op	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	No					
	Mult2	No					

Register result status:

Clock		<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
1	FU				Load1					

Cycle-2

Instruction status:

				Issue	Exec	Write
Instruction		<i>j</i>	<i>k</i>		Comp	Result
LD	F6	34+	R2	1		
LD	F2	45+	R3	2		
MULTD	F0	F2	F4			
SUBD	F8	F6	F2			
DIVD	F10	F0	F6			
ADDD	F6	F8	F2			

	Busy	Address
Load1	Yes	34+R2
Load2	Yes	45+R3
Load3	No	

Reservation Stations:

on Stations:

Time	Name	Busy	Op	S1 Vj	S2 Vk	RS Qj	RS Qk
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	No					
	Mult2	No					

Register result status:

Clock	<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
2		Load2							
	<i>FU</i>			Load1					

Cycle-3

Instruction status:

Instruction		<i>j</i>	<i>k</i>	<i>Issue</i>	<i>Comp</i>	<i>Result</i>		<i>Busy</i>	<i>Address</i>
LD	F6	34+	R2	1	3		Load1	Yes	34+R2
LD	F2	45+	R3	2			Load2	Yes	45+R3
MULTD	F0	F2	F4	3			Load3	No	
SUBD	F8	F6	F2						
DIVD	F10	F0	F6						
ADDD	F6	F8	F2						

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	Yes	MULTD		R(F4)	Load2	
	Mult2	No					

Register result status:

Clock		F0	F2	F4	F6	F8	F10	F12	...	F30
3	FU	Mult1	Load2		Load1					

Cycle-4

Instruction status:

				Issue	Exec Comp	Write Result		
Instruction		<i>j</i>	<i>k</i>				Busy	Address
LD	F6	34+	R2	1	3	4	Load1	No
LD	F2	45+	R3	2	4		Load2	Yes 45+R3
MULTD	F0	F2	F4	3			Load3	No
SUBD	F8	F6	F2	4				
DIVD	F10	F0	F6					
ADDD	F6	F8	F2					

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
Time	Name	Busy	Op	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
	Add1	Yes	SUBD	M(A1)			Load2
	Add2	No					
	Add3	No					
	Mult1	Yes	MULTD		R(F4)	Load2	
	Mult2	No					

Register result status:

Clock		<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
4	FU	Mult1	Load2		M(A1)	Add1				

Cycle-5

Instruction status:

				Issue	Exec Comp	Write Result		
Instruction		<i>j</i>	<i>k</i>				Busy	Address
LD	F6	34+	R2	1	3	4	Load1	No
LD	F2	45+	R3	2	4	5	Load2	No
MULTD	F0	F2	F4	3			Load3	No
SUBD	F8	F6	F2	4				
DIVD	F10	F0	F6	5				
ADDD	F6	F8	F2					

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
Time	Name	Busy	Op	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
2	Add1	Yes	SUBD	M(A1)	M(A2)		
	Add2	No					
	Add3	No					
10	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
5	FU	Mult1	M(A2)		M(A1)	Add1	Mult2			

Cycle-6

Instruction status:

				Issue	Exec Comp	Write Result		
Instruction		<i>j</i>	<i>k</i>				Busy	Address
LD	F6	34+	R2	1	3	4	Load1	No
LD	F2	45+	R3	2	4	5	Load2	No
MULTD	F0	F2	F4	3			Load3	No
SUBD	F8	F6	F2	4				
DIVD	F10	F0	F6	5				
ADDD	F6	F8	F2	6				

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
Time	Name	Busy	Op	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
1	Add1	Yes	SUBD	M(A1)	M(A2)		
	Add2	Yes	ADDD		M(A2)	Add1	
	Add3	No					
9	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock	<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
6									
FU	Mult1	M(A2)		Add2	Add1	Mult2			

Cycle-7

Instruction status:

				Issue	Exec Comp	Write Result		
Instruction		<i>j</i>	<i>k</i>				Busy	Address
LD	F6	34+	R2	1	3	4	Load1	No
LD	F2	45+	R3	2	4	5	Load2	No
MULTD	F0	F2	F4	3			Load3	No
SUBD	F8	F6	F2	4	7			
DIVD	F10	F0	F6	5				
ADDD	F6	F8	F2	6				

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
Time	Name	Busy	Op	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
0	Add1	Yes	SUBD	M(A1)	M(A2)		
	Add2	Yes	ADDD		M(A2)	Add1	
	Add3	No					
8	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock	<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
7	FU								
	Mult1	M(A2)		Add2	Add1	Mult2			

Cycle-8

Instruction status:

				Exec	Write		
Instruction	<i>j</i>	<i>k</i>	Issue	Comp	Result	Busy	Address
LD	F6	34+	R2	1	3	4	Load1
LD	F2	45+	R3	2	4	5	Load2
MULTD	F0	F2	F4	3			Load3
SUBD	F8	F6	F2	4	7	8	
DIVD	F10	F0	F6	5			
ADDD	F6	F8	F2	6			

Reservation Stations:

				S1	S2	RS	RS
Time	Name	Busy	Op	Vj	Vk	Qj	Qk
	Add1	No					
2	Add2	Yes	ADDD	(M-M)	M(A2)		
	Add3	No					
7	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock	F0	F2	F4	F6	F8	F10	F12	...	F30
8	Mult1	M(A2)		Add2	(M-M)	Mult2			

Cycle-9

Instruction status:

Instruction	<i>j</i>	<i>k</i>	<i>Issue</i>	<i>Exec Comp</i>	<i>Write Result</i>		<i>Busy</i>	<i>Address</i>
LD	F6	34+	R2	1	3	4	Load1	No
LD	F2	45+	R3	2	4	5	Load2	No
MULTD	F0	F2	F4	3			Load3	No
SUBD	F8	F6	F2	4	7	8		
DIVD	F10	F0	F6	5				
ADDD	F6	F8	F2	6				

Reservation Stations:

<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>S1 Vj</i>	<i>S2 Vk</i>	<i>RS Qj</i>	<i>RS Qk</i>
	Add1	No					
1	Add2	Yes	ADDD	(M-M)	M(A2)		
	Add3	No					
6	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock	<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
9	FU Mult1 M(A2) Add2 (M-M) Mult2								

Cycle-10

Instruction status:

Instruction		<i>j</i>	<i>k</i>	<i>Issue</i>	<i>Exec Comp</i>	<i>Write Result</i>		<i>Busy</i>	<i>Address</i>
LD	F6	34+	R2	1	3	4	Load1	No	
LD	F2	45+	R3	2	4	5	Load2	No	
MULTD	F0	F2	F4	3			Load3	No	
SUBD	F8	F6	F2	4	7	8			
DIVD	F10	F0	F6	5					
ADDD	F6	F8	F2	6	10				

Reservation Stations:

<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>S1 Vj</i>	<i>S2 Vk</i>	<i>RS Qj</i>	<i>RS Qk</i>
	Add1	No					
0	Add2	Yes	ADDD	(M-M)	M(A2)		
	Add3	No					
5	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
10	FU	Mult1	M(A2)		Add2	(M-M)	Mult2			

Cycle-11

Instruction status:

Instruction		<i>j</i>	<i>k</i>	<i>Issue</i>	<i>Exec Comp</i>	<i>Write Result</i>		<i>Busy</i>	<i>Address</i>
LD	F6	34+	R2	1	3	4	Load1	No	
LD	F2	45+	R3	2	4	5	Load2	No	
MULTD	F0	F2	F4	3			Load3	No	
SUBD	F8	F6	F2	4	7	8			
DIVD	F10	F0	F6	5					
ADDD	F6	F8	F2	6	10	11			

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
<i>Time</i>	<i>Name</i>	<i>Busy</i>	<i>Op</i>	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
	Add1	No					
	Add2	No					
	Add3	No					
4	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
11	FU	Mult1	M(A2)		(M-M+M(M-M))	Mult2				

Cycle-12

Instruction status:

				Issue	Exec Comp	Write Result		
Instruction		<i>j</i>	<i>k</i>				Busy	Address
LD	F6	34+	R2	1	3	4	Load1	No
LD	F2	45+	R3	2	4	5	Load2	No
MULTD	F0	F2	F4	3			Load3	No
SUBD	F8	F6	F2	4	7	8		
DIVD	F10	F0	F6	5				
ADDD	F6	F8	F2	6	10	11		

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
Time	Name	Busy	Op	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
	Add1	No					
	Add2	No					
	Add3	No					
3	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
12	FU	Mult1	M(A2)		(M-M+M(M-M)	Mult2				

Cycle-13

Instruction status:

				Issue	Exec Comp	Write Result		
Instruction		<i>j</i>	<i>k</i>				Busy	Address
LD	F6	34+	R2	1	3	4	Load1	No
LD	F2	45+	R3	2	4	5	Load2	No
MULTD	F0	F2	F4	3			Load3	No
SUBD	F8	F6	F2	4	7	8		
DIVD	F10	F0	F6	5				
ADDD	F6	F8	F2	6	10	11		

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
Time	Name	Busy	Op	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
	Add1	No					
	Add2	No					
	Add3	No					
2	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock		<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
13	FU	Mult1	M(A2)		(M-M+M(M-M)	Mult2				

Cycle-14

Instruction status:

				Issue	Exec Comp	Write Result		
Instruction		<i>j</i>	<i>k</i>				Busy	Address
LD	F6	34+	R2	1	3	4	Load1	No
LD	F2	45+	R3	2	4	5	Load2	No
MULTD	F0	F2	F4	3			Load3	No
SUBD	F8	F6	F2	4	7	8		
DIVD	F10	F0	F6	5				
ADDD	F6	F8	F2	6	10	11		

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
Time	Name	Busy	Op	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
	Add1	No					
	Add2	No					
	Add3	No					
1	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock	<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
14	Mult1	M(A2)		(M-M+M(M-M)	Mult2				

Cycle-15

Instruction status:

				Exec		Write		
Instruction		<i>j</i>	<i>k</i>	<i>Issue</i>	<i>Comp</i>	<i>Result</i>	Busy	Address
LD	F6	34+	R2	1	3	4	Load1	No
LD	F2	45+	R3	2	4	5	Load2	No
MULTD	F0	F2	F4	3	15		Load3	No
SUBD	F8	F6	F2	4	7	8		
DIVD	F10	F0	F6	5				
ADDD	F6	F8	F2	6	10	11		

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
Time	Name	Busy	Op	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
	Add1	No					
	Add2	No					
	Add3	No					
0	Mult1	Yes	MULTD	M(A2)	R(F4)		
	Mult2	Yes	DIVD		M(A1)	Mult1	

Register result status:

Clock	<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
15	FU	Mult1	M(A2)		(M-M+M(M-M)	Mult2			

Cycle-16

Instruction status:

				Exec		Write		
Instruction		<i>j</i>	<i>k</i>	<i>Issue</i>	<i>Comp</i>	<i>Result</i>	Busy	Address
LD	F6	34+	R2	1	3	4	Load1	No
LD	F2	45+	R3	2	4	5	Load2	No
MULTD	F0	F2	F4	3	15	16	Load3	No
SUBD	F8	F6	F2	4	7	8		
DIVD	F10	F0	F6	5				
ADDD	F6	F8	F2	6	10	11		

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
Time	Name	Busy	Op	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	No					
40	Mult2	Yes	DIVD	M*F4	M(A1)		

Register result status:

Clock	<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
16	FU	M*F4	M(A2)		(M-M+M(M-M)	Mult2			

Cycle-55

Instruction status:

				<i>Exec</i>		<i>Write</i>		
Instruction		<i>j</i>	<i>k</i>	<i>Issue</i>	<i>Comp</i>	<i>Result</i>		
LD	F6	34+	R2	1	3	4	Load1	Busy
LD	F2	45+	R3	2	4	5	Load2	Address
MULTD	F0	F2	F4	3	15	16	Load3	
SUBD	F8	F6	F2	4	7	8		
DIVD	F10	F0	F6	5				
ADDD	F6	F8	F2	6	10	11		

Reservation Stations:

				<i>S1</i>	<i>S2</i>	<i>RS</i>	<i>RS</i>
Time	Name	Busy	Op	<i>Vj</i>	<i>Vk</i>	<i>Qj</i>	<i>Qk</i>
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	No					
1	Mult2	Yes	DIVD	M*F4	M(A1)		

Register result status:

Clock		<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
55	FU	M*F4	M(A2)		(M-M+M(M-M)	Mult2				

Cycle-56

Instruction status:

Instruction		<i>j</i>	<i>k</i>	<i>Issue</i>	<i>Exec Comp</i>	<i>Write Result</i>		<i>Busy</i>	<i>Address</i>
LD	F6	34+	R2	1	3	4	Load1	No	
LD	F2	45+	R3	2	4	5	Load2	No	
MULTD	F0	F2	F4	3	15	16	Load3	No	
SUBD	F8	F6	F2	4	7	8			
DIVD	F10	F0	F6	5	56				
ADDD	F6	F8	F2	6	10	11			

Reservation Stations:

on Stations:

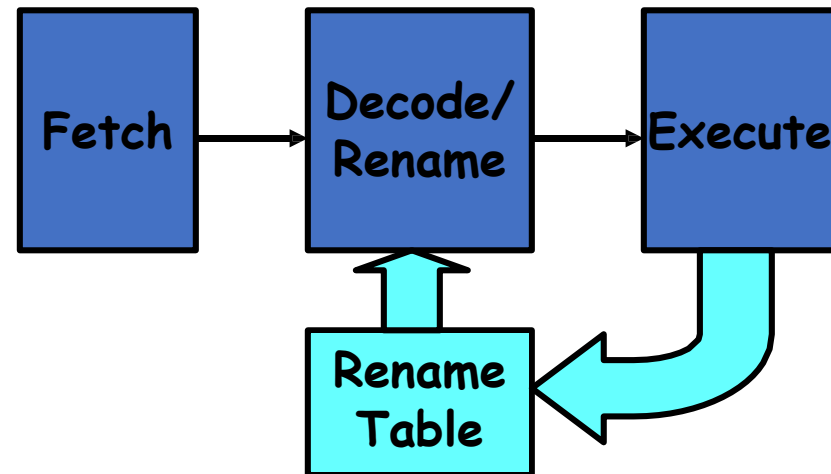
				S1	S2	RS	RS
Time	Name	Busy	Op	Vj	Vk	Qj	Qk
	Add1	No					
	Add2	No					
	Add3	No					
	Mult1	No					
0	Mult2	Yes	DIVD	M*F4	M(A1)		

Register result status:

Clock		<i>F0</i>	<i>F2</i>	<i>F4</i>	<i>F6</i>	<i>F8</i>	<i>F10</i>	<i>F12</i>	...	<i>F30</i>
56	FU	M*F4	M(A2)		(M-M+M(M-M)	Mult2				

Explicit Register Renaming

- Make use of a *physical* register file that is larger than number of registers specified by ISA
- Keep a translation table:
 - ISA register => physical register mapping
 - Physical register becomes free when not being used by any instructions in progress.



Explicit Register Renaming

- Rapid access to a table of translations
- A physical register file that has more registers than specified by the ISA
- Many modern architectures use explicit register renaming + Tomasulo-like reservation stations to control execution.

Summary

- WAR and WAW hazards are primarily eliminated by explicit register renaming. It cannot mitigate RAW hazards. However, it will still rename it so that the processor moves further. Once it runs out of all the registers, it will stall.
- Implicit register renaming through reservation stations also mitigates WAR and WAW hazards. For RAW hazards, the processor will wait and stall.